Firm Automation and Cost of Lending: Micro Insights for an Emerging Market*

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Abstract

This paper studies whether firm automation is reflected in the cost of bank financing. We link customs records on automation-related capital goods imports to monthly credit registry data and construct a panel of approximately 2.4 million bank-firm-month credit observations for Türkiye over 2018–2025. Automated firms borrow at significantly lower interest rates and obtain larger loan volumes. The interest-rate discount remains robust after matching firms on pre-sample observables and in an event-study design around the release of ChatGPT-4, which shifted the value of automation-complementary technologies. The discount is reversed at state-owned banks and larger at banks with greater ICT-related human capital, consistent with stronger pricing of automation where lenders have greater incentives and capacity to assess borrower risk. On the firm side, the discount is larger for firms with high income volatility, stronger bank relationships, and greater agency conflicts. It also widens during episodes of unexpected monetary tightening and is amplified when automation is combined with AI-related computing investment. Automated firms also perform better on several real outcomes, including sales, employment, exports, and capital investment. Taken together, the evidence suggests that automation shapes firm outcomes not only through production, but also through access to external finance.

JEL classification: G21; O33; G32

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1 Introduction

Automation is changing production in manufacturing. Industrial robots and programmable production systems are increasingly common, and recent advances in artificial intelligence have broadened the set of tasks that can be automated. A large literature documents the consequences of this transformation for employment, wages, and productivity (Acemoglu et al., 2020; Bonfiglioli et al., 2024). However, the financial side of automation remains less well understood. In particular, it is unclear whether banks view automation as altering borrower risk, and whether they incorporate this assessment into loan pricing. This question matters especially in bank-dependent economies, where loan pricing directly affects firm investment and growth. If banks price automation, financing conditions may become an additional channel through which technology adoption affects firm outcomes.

The effect of automation on borrowing costs is theoretically ambiguous. By replacing variable labor inputs with capital whose cost structure is largely fixed and predictable, automation stabilizes cash flows and reduces the variance of earnings, lowering the default probability that lenders must price (Minton and Schrand, 1999). Automation investment is also observable and costly to reverse: capital goods purchases recorded at the customs border credibly reveal organizational quality and commit internal resources to productive use, mitigating the information asymmetries at the heart of credit market frictions (Myers and Majluf, 1984). At the same time, automation requires large up-front outlays that are often debt-financed, raises operating leverage, and exposes the firm to technology-related risks, all of which could lead lenders to charge more rather than less. Whether banks price these opposing forces is ultimately an empirical question. It has been difficult to answer because it requires data that jointly observe technology adoption and loan terms within individual bank-firm relationships.

We provide micro-level evidence on how firm automation is associated with the cost of bank lending. We assemble a dataset linking four administrative registries for Türkiye: transaction-level customs records covering the universe of Turkish firms' imports, the credit registry capturing all commercial lending relationships between domestic banks

and firms at monthly frequency, firm financial statements from the tax authority, and employee-level occupational records from the social security administration. Following Acemoglu et al. (2020), we measure automation from imports of automation-related capital goods (e.g., industrial robots, computer-numerically-controlled (CNC) machines, welding automation equipment, programmable logic controllers, and automated guided vehicles, among others) identified through a curated set of six-digit Harmonized System codes. A firm is classified as automated in a given month if the trailing 24-month sum of such imports is positive, a definition that captures medium-run automation status rather than one-off transactions. The resulting panel comprises approximately 2.4 million bank-firm-month observations, covering credit relationships between 48 banks and 73,683 manufacturing firms from January 2018 to December 2025.

Türkiye provides an ideal setting for this analysis. Bank credit is the dominant source of external finance for Turkish firms, so loan pricing has immediate real consequences. The manufacturing sector has been undergoing rapid technological catch-up, generating substantial variation in adoption across firms and over time. And the sample period spans extraordinary macro-financial variation including a tightening cycle that took the policy rate to 50% over 2023–2024, which allows us to examine how the pricing of borrower technology behaves across monetary conditions. The setting is also relevant for other middle-income economies where manufacturing firms rely heavily on bank credit while adopting frontier technologies.

Our baseline specification regresses the weighted-average interest rate on Turkish lira loans at the bank-firm-month level on the automation indicator, absorbing firm, bank-by-time, and industry-by-time fixed effects. This three-way structure imposes demanding identification requirements: the automation coefficient is estimated from firms that transition into or out of automated status, relative to other firms borrowing from the same bank in the same month and operating in the same sector in the same month. Time-invariant firm heterogeneity, monthly bank-level credit supply conditions, and industry-level demand cycles are thereby purged simultaneously. We find that automated firms borrow at interest rates approximately 8 to 9 basis points lower than their non-automated

counterparts, and that they also secure significantly larger loan volumes. The average effect is modest in absolute terms, but it is estimated within firm and remains stable across demanding specifications. Moreover, the average estimate conceals considerably larger effects in well-defined states of the world: the discount is roughly 31 basis points at private banks, widens by about 30 basis points after a major shock to automation-complementary technologies (i.e., the introduction of ChatGPT-4), and deepens by a further 24 basis points when automation is combined with firm-level AI investment.

Two concerns regarding the causal interpretation of our results are non-random selection into automation and confounding firm dynamics. We address selection by re-estimating the model on a matched sample constructed via one-to-one multivariate distance matching on pre-sample (2017) firm characteristics, with exact matching on sector and regulatory size class; the estimates are broadly unchanged. We address dynamics by exploiting the public release of ChatGPT-4 in March 2023 as an exogenous, globally unanticipated shock to the value of automation-complementary technologies (Noy and Zhang, 2023; Brynjolfsson et al., 2025). Firms with automation infrastructure already in place were better positioned to benefit from the advance in AI capability, and lenders aware of this complementarity should reprice their loans accordingly. In a difference-in-differences design on the matched sample, the automation level effect is statistically indistinguishable from zero while the post-ChatGPT-4 interaction is negative, highly significant, and economically large (29 to 31 basis points). An event study shows coefficients hovering around zero throughout the ten quarters preceding the release before declining sharply and persistently thereafter. This pattern is less consistent with slow-moving differences in firm quality and more consistent with banks updating their assessment of automated borrowers after the technology environment changed.

We next examine where the automation discount is strongest. On the lender side, the discount is more than fully reversed at state-owned banks, whose lending decisions are known to deviate from profit-maximizing benchmarks under soft budget constraints and political objectives (Kornai et al., 2003; Sapienza, 2004; Dinç, 2005; Bertay et al., 2015). This pattern is consistent with the interpretation that the discount reflects risk-based pricing:

it is concentrated among lenders for whom commercial pricing incentives are likely to be stronger. The discount is larger at banks with a higher share of ICT-skilled employees, suggesting that banks with stronger technological capacity may be better able to interpret automation-related information (Pierri and Timmer, 2022). On the borrower side, the discount is concentrated among firms for which automation is likely to be more informative. It is larger for firms 1) with high sales volatility, consistent with an earnings stability channel in which the marginal value of automation-induced cash flow smoothing is greatest where baseline uncertainty is highest (Minton and Schrand, 1999), 2) that concentrate their borrowing with a given bank, consistent with an information channel in which relationship lenders are best positioned to evaluate the risk implications of borrower investments (Bolton et al., 2016), 3) whose financial profile, high free cash flow combined with low asset growth, signals elevated agency conflicts, consistent with automation serving as a credible commitment of slack resources to productive use (Jensen, 1986; Myers and Majluf, 1984).

We complement the main credit-pricing results with additional evidence on realized credit risk, monetary-policy conditions, technology complementarities, and firm performance. First, automated firms' loans are significantly less likely to be classified as nonperforming, indicating that the lower rates they receive reflect genuine reductions in realized credit risk rather than mispricing. Second, the automation discount widens during episodes of unexpected monetary tightening — identified from policy rate surprises relative to survey expectations — consistent with the bank lending channel: when funding becomes costly and risk appetite contracts, lenders discriminate more sharply across borrowers, and the risk signal embedded in automation becomes more valuable (Jiménez et al., 2012). Third, physical automation and AI adoption are complements in credit pricing: while hardware-based AI investment alone has a marginal effect, its interaction with automation is negative and highly significant, indicating that the combined effect exceeds the sum of the parts (Babina et al., 2024; Brynjolfsson et al., 2025). Finally, automated firms perform better on several real outcomes, including sales, employment, exports, and capital investment. These patterns suggest that the lending discount is related to observable

differences in firm performance rather than pricing differences alone.

Our paper contributes to three strands of literature. First, we contribute to the literature on automation and firm performance. Existing work documents the effects of robot adoption on employment, wages, and productivity at the aggregate and firm levels (Acemoglu et al., 2020; Bonfiglioli et al., 2024; Bessen, 2019). We add a financing dimension to this literature. We show that automation is reflected in loan pricing, suggesting that technology adoption may affect firms not only through production, but also through access to external finance. To our knowledge, this is the first paper to link physical automation, measured from capital goods imports rather than digitalization indices, to loan-level lending terms.

Second, we contribute to the literature on information frictions and the pricing of bank credit. A long tradition studies how lenders overcome information asymmetries, from delegated monitoring and credit rationing to relationship lending (Petersen and Rajan, 1994; Bolton et al., 2016) and the transmission of credit supply shocks (Khwaja and Mian, 2008; Jiménez et al., 2012). We introduce a new, verifiable borrower attribute into this framework: technology investment recorded in administrative customs data. We show that lenders price this hard information, that its transmission is strongest within deep lending relationships, and that it interacts systematically with lender ownership and lender technology — connecting the borrower-side information literature to work on state banks and credit allocation (Sapienza, 2004; Dinç, 2005; Bertay et al., 2015).

Third, we contribute to an emerging literature on technology in finance and the economics of AI. Prior work shows that banks' own technology improves their resilience and lending capacity (Pierri and Timmer, 2022) and that firm-level AI investment raises growth and innovation (Babina et al., 2024; Noy and Zhang, 2023; Brynjolfsson et al., 2025). Our results point to technology on both sides of the credit relationship. Borrower automation is priced most aggressively by technology-intensive banks, and borrower-side automation and AI investments are complements in reducing lending costs. The ChatGPT-4 event study provides, to our knowledge, some of the first credit-market evidence on how a generative-AI capability shock propagates to loan pricing.

The rest of the paper is organized as follows. Section 2 discusses the datasets and sample formation process. Section 3 gives details on the empirical setting. Section 4 presents baseline results as well as alternative empirical designs, heterogeneity analyses, the results pertaining to other lending outcomes, robustness checks, the role of AI adoption and macroeconomic effects. Section 5 presents concluding remarks.

2 Data

In our empirical analysis, we employ different administrative datasets, which are made available by the Central Bank of the Republic of Türkiye (CBRT), to analyze how the degree of firm automation is reflected in credit pricing.

First, we use the dataset consisting of transaction-level foreign trade records maintained by the Ministry of Trade. This database covers the entire universe of Turkish firms' exports and imports together with information on transaction value, currency of denomination, destination/source country, and (most importantly for our case) the 6-digit Harmonized System (HS) codes describing the classification of exported/imported products and services (Akgündüz and Fendoğlu, 2022). Each record is linked to a unique firm identifier enabling matching with other administrative registries. In our work, we draw on this dataset to identify firm-level imports of automation-related capital goods based on a curated set of HS product codes, the construction of which is described in detail in Section 3. The firms engaging with such imports are deemed to have adopted a higher degree of automation in their production and business operations.

Second, we rely on the Credit Registry maintained by the Banking Regulation and Supervision Agency (BRSA). This dataset captures all commercial credit relationships between domestic banks and firms, providing the outstanding loan stock, the quoted interest rate, the currency of denomination, the original maturity, and the default status of the borrower. The weighted-average interest rate on Turkish lira-denominated loans serves as the primary outcome variable in our baseline estimations. In extended analyses, we also examine loan stock, maturity and non-performing loan ratio in order to assess whether automation shapes not only the price but also non-price terms and credit risk of lending

relationships.

Third, we collect annual firm financial statements retrieved from the Revenue Administration, prepared in compliance with Tax Procedure Law and processed jointly by Turkish Statistical Institute (TurkStat) and CBRT. These records equip us with balance sheet and income statement information for incorporated firms in Türkiye (the ones subject to corporate income tax reporting requirements). We exploit this database to follow firm features (such as regulatory firm size classifications and NACE sector information) and to construct the time-varying firm-level control variables used throughout the empirical analysis – namely firm assets, profitability, leverage, liquidity and tangibility. Moreover, we complement the financial information with employee-level data from the Social Security Institution, which reports the occupational composition of each firm's workforce at ISCO-08 code resolution. This registry provides firm-level employment counts used in the subsequent real-side analysis as well as forms the basis for the employment-based automation proxy used in the robustness checks.

Our sample covers the period from January 2018 to December 2025. To perform sample composition, we begin with the customs records and identify, for each firm and month, the value of imports falling under the pre-determined set of 6-digit HS codes associated with automation-related capital goods. These transaction-level figures are then aggregated to the firm-month level and merged with credit data in a subsequent step. Turning to the credit database, we drop observations with zero reported interest rates or zero outstanding loan stock, as these likely reflect irregularities or dormant credit relationships rather than active lending dynamics. The remaining records are collapsed to the bank-firm-month level and merged with the firm-automation data constructed in the first step. Finally, we incorporate the financial statement and employment data while restricting the sample to manufacturing firms – for whom imports of capital goods provide a meaningful and economically relevant measure of automation – and excluding firms with fewer than two employees. Ultimately, our sample comprises approximately 2.4 million observations, corresponding to credit relationships between 48 banks and 73,683 manufacturing firms. The summary statistics are presented in Table 1, while the variable definitions are given

in Tables A1 and A2 of the Appendix.

3 Empirical Design

This section describes the empirical design used to estimate the relationship between firm automation and the cost of lending. Following the methodology of Acemoglu et al. (2020), we construct a firm-level binary measure of automation intensity using transaction-level customs records. Specifically, we identify whether sample firms have imported capital goods related to automation-related equipment including industrial robots, CNC machines, welding automation equipment, programmable logic controllers, and automated guided vehicles, among others.[1] The *Automation* indicator takes the value of one if the 24-month trailing sum of a firm's imports under these HS codes is positive, and zero otherwise. This trailing-sum approach captures the medium-run automation status of a firm rather than reflecting one-off or transitory import transactions.[2]

Our baseline specification takes the following form:

$$\text{Int Rate}_{f,b,t} = \beta\, Automation_{f,t} + \mathbf{X}'_{f,t}\boldsymbol{\gamma} + \alpha_f + \delta_{b,t} + \lambda_{s,t} + \varepsilon_{f,b,t} \tag{1}$$

where $IntRate_{f,b,t}$ denotes the weighted average interest rate on loans extended by bank b to firm f in month t. The coefficient of interest, β, captures the within-firm effect of automation on the cost of lending. $\mathbf{X}_{f,t}$ is a vector of time-varying firm-level controls comprising firm size (log-transformed total assets), return on assets, leverage (debt-to-assets), the current ratio, and asset tangibility. The terms α_f, $\delta_{b,t}$, and $\lambda_{s,t}$ denote firm, bank-by-time, and industry-by-time fixed effects, respectively. $\varepsilon_{f,b,t}$ is the idiosyncratic error term. Standard errors are clustered at the firm level.

The three-way fixed effect structure provides a demanding identification framework. The firm fixed effect α_f absorbs all time-invariant firm characteristics — including sector

[1] The full list of 6-digit HS codes used to identify automation-related capital goods is as follows: 847950, 847989, 845600–846699, 846820–846899, 851511–851519, 851521, 851531, 851580, 851590, 844600–844699, 844700–844799, 844400–844590, 842831–842839, 903200–903299, and 847780.

[2] As discussed in Section 4.6, the results are robust to alternative window lengths of 12 and 36 months, as well as to a one-shift specification in which the indicator switches permanently to one upon the firm's first automation-related import.

affiliation, ownership structure, management quality, and geographic location — so that identification comes entirely from within-firm variation over time. The bank-by-time fixed effect $\delta_{b,t}$ controls for all bank-specific factors that vary at monthly frequency, such as funding costs, capital adequacy, risk appetite, and loan growth strategies, ensuring that we compare firms borrowing from the same bank in the same month. The industry-by-time fixed effect $\lambda_{s,t}$ absorbs sector-level credit demand cycles, so that the automation coefficient reflects firm-level variation within the same sector-month cell. Taken together, β is identified from firms that transition into — or out of — an automated state relative to other firms borrowing from the same bank in the same month and operating in the same sector in the same month, providing a particularly stringent test of the automation-lending cost relationship.

4 Empirical Findings

4.1 Baseline Results

Table 2 presents the baseline estimates. The coefficient on *Automation* is negative and statistically significant at the 1% level in both columns (1) and (2), indicating that automated firms access bank credit at an interest rate approximately 8 to 9 basis points lower relative to their non-automated counterparts. While modest in absolute terms, this estimate is identified from within-firm variation in a highly saturated model that absorbs firm-level time-invariant heterogeneity, bank-specific credit supply conditions at monthly frequency, and sector-level demand dynamics in a simultaneous manner. Given this demanding specification, the stability of the estimate across columns is informative. Notably, even modest reductions in borrowing costs carry well-documented consequences in bank-dependent settings (Khwaja and Mian, 2008; Jiménez et al., 2012). Columns (3) and (4) show that automated firms also secure larger credit balances, with the coefficient on *Automation* remaining positive and statistically significant across both specifications. These results suggest that automation improves firms' access to bank financing along both price and quantity dimensions, consistent with the broader evidence that technol-

ogy adoption enhances operational performance in ways reflected in external financing conditions (Acemoglu et al., 2020; Bonfiglioli et al., 2024).

4.2 Alternative Empirical Design

In this subsection, we follow alternative empirical designs to ensure the internal validity of our main findings. First, we construct a matched sample and re-estimate the model to address the concern that automated and non-automated firms may differ along pre-existing observable dimensions. Firms are divided into two groups: those whose *Automation* indicator takes the value of one at any point during the 2018–2025 period, and those for whom it remains zero throughout. We apply 1-to-1 multivariate distance matching without replacement based on pre-sample (2017) firm characteristics — profitability, liquidity, total assets, tangibility, and leverage — with exact matching imposed on 2-digit NACE sector classifications and regulatory firm size categories, yielding 3,726 treated firms matched to 3,726 control firms.

Columns (1) and (2) of Table 3 report the baseline estimates on the matched sample. The coefficient on *Automation* remains negative and statistically significant, broadly in line with the full-sample estimates in Table 2, suggesting that the main finding is not an artifact of pre-existing observable differences between automated and non-automated firms.

As a complementary identification strategy, we exploit the widespread introduction of ChatGPT-4 in March 2023 as an exogenous shock to automation-complementary technologies. The release of ChatGPT-4 represented a discrete and globally unanticipated discontinuity in the frontier of AI capability, with well-documented effects on firm-level productivity and task performance (Noy and Zhang, 2023; Brynjolfsson et al., 2025). Firms already equipped with automation infrastructure are better positioned to exploit such AI advances, potentially amplifying their informational and risk advantages in the eyes of lenders. We implement this as a difference-in-differences specification on the matched sample:

$$\begin{aligned}\text{Int Rate}_{f,b,t} = \beta_1\, Automation_{f,t} + \beta_2 \left(Automation_{f,t} \times Post\ ChatGPT\text{-}4_t\right) \\ + \mathbf{X}'_{f,t}\boldsymbol{\gamma} + \alpha_f + \delta_{b,t} + \lambda_{s,t} + \varepsilon_{f,b,t}\end{aligned} \tag{2}$$

where *Post ChatGPT-4*$_t$ takes the value of one from March 2023 onwards, and β_2 is the coefficient of interest, capturing the differential change in lending costs for automated firms following the event. Columns (3) and (4) of Table 3 report the results on the matched sample. The coefficient on *Automation* alone is statistically indistinguishable from zero, while *Automation* × *Post ChatGPT-4* is negative and highly significant, indicating that the lending cost advantage of automated firms widened considerably after the ChatGPT-4 introduction.

Figure 1 provides further support through an event study, where the *Automation* indicator is interacted with relative-time dummies defined at quarterly frequency over a [-10, +10] quarter window, with the model estimated at monthly frequency. The pre-event coefficients are generally close to zero and do not display a clear downward trend before the event, supporting the parallel trends assumption, before declining sharply and persistently following the March 2023 event date.

4.3 Heterogeneous Effects Across Bank Characteristics

We next examine whether the automation discount varies with bank characteristics. Table 4 examines whether the baseline effect varies across bank types and technological capabilities. Column (1) introduces an interaction between *Automation* and a binary indicator for state-owned banks, while column (2) interacts *Automation* with the share of ICT-related employees in the bank's total workforce.[3]

The results in column (1) reveal a striking asymmetry. While the coefficient on *Automation* remains negative and significant at private banks, the interaction *Automation* ×

[3]We use ISCO-4 codes including 2511, 2523, 3511 and 3512 to measure ICT-related labor share in banks' total employment. The occupation codes correspond to system analysts, computer network professionals, ICT operations technicians and ICT user support technicians, respectively. These occupations are closely associated with strategic design and implementation of the systems and networks that enable banks to screen automated firms more effectively (Haskel and Westlake, 2017), and with the operational continuity that helps ICT-intensive banks distinguish productive automated firms from others (Bessen, 2019).

State Bank is positive and highly significant, more than offsetting the baseline effect. This finding is in line with the view that state-owned banks, operating under soft budget constraints and subject to non-commercial lending objectives, are less likely to translate borrower risk signals into pricing decisions (Kornai et al., 2003; Sapienza, 2004). State banks' lending behaviour is known to deviate systematically from profit-maximising benchmarks, driven by political and social objectives that override risk-based pricing considerations (Dinç, 2005; Bertay et al., 2015). The absence of a comparable discount at state-owned banks suggests that the baseline result is not a mechanical correlation between automation and lower interest rates, but is more likely to arise when banks have stronger incentives to price borrower risk.

Turning to column (2), the interaction between *Automation* and *Bank ICT Emp* is negative and statistically significant, indicating that banks with a larger share of ICT-skilled employees translate the automation signal into larger reductions in lending costs. This result accords with the broader evidence that technologically sophisticated lenders exhibit superior screening and monitoring capabilities, which improve the quality of their credit assessments (Petersen and Rajan, 1994; Pierri and Timmer, 2022).

4.4 Heterogeneous Effects Across Firm Characteristics

We then examine whether the automation discount varies with firm characteristics. Table 5 investigates whether the baseline effect varies across firm-level characteristics that are theoretically linked to the channels through which automation is expected to reduce the cost of lending. Columns (1), (2), and (3) interact *Automation* with *Sales Volatility*, *Loan Density*, and *Agency Conflicts*, respectively.

The interaction between *Automation* and *Sales Volatility* in column (1) is negative and statistically significant, indicating that the automation discount is more pronounced for firms with greater operational income volatility. This result reinforces the applicability of earnings stability channel. By replacing variable labor inputs with more predictable capital expenditures, automation stabilizes the firm's cost structure and reduces the variance of cash flows. The marginal value of this stabilization is greatest for firms whose income

streams are inherently more uncertain, as lenders embed larger uncertainty premia in credit spreads when future cash flows are more difficult to forecast (Minton and Schrand, 1999). The finding therefore suggests that automation functions as a credible commitment to reducing operational risk, with the effect amplified where baseline earnings uncertainty is highest.

Column (2) shows that the interaction between *Automation* and *Loan Density* — the share of a given bank in the firm's total TL credit borrowing — is negative and statistically significant. Firms that concentrate their borrowing with a specific bank benefit more from automation in terms of lending cost reductions. information channel. In the context of information channel, relationship banks accumulate private information about borrowers over time and are better positioned to evaluate and price the risk implications of firm-level investments such as automation (Bolton et al., 2016). The finding suggests that the information value of automation is most effectively transmitted in settings where the lender has developed a deeper understanding of the borrower's operational characteristics.

Column (3) introduces the interaction between *Automation* and *Agency Conflicts*, a binary indicator for firms simultaneously exhibiting above-median free cash flow to assets and below-median asset growth. This profile captures firms generating internal resources without reinvesting them productively — a configuration associated with elevated managerial discretion and information asymmetry between managers and creditors (Jensen, 1986). The interaction coefficient is negative and highly significant, implying that the automation discount is markedly stronger for firms whose financial profile raises agency concerns. Automation credibly addresses these concerns by committing slack resources to productive capital investment, thereby constraining managerial discretion and improving the transparency of the firm's operational plans to outside lenders (Myers and Majluf, 1984). The result lends direct support to the information asymmetry channel and suggests that automation investment carries a distinct signaling value precisely where lenders face the greatest uncertainty about the deployment of internally generated funds.

4.5 Monetary Policy Shocks

We next examine whether monetary conditions affect the automation discount. In Table 6, the key variable of interest, *Automation* × *MP Shock*, interacts the automation indicator with the monetary policy shock, defined as the residual from a regression of the difference between the actual policy rate and market expectations — sourced from the CBRT Survey of Market Participants — on a set of macroeconomic predictors, including 12-month forward inflation expectation, annual change in industrial production index, annual credit growth and annual change in Istanbul Stock Exchange (BIST100) index. A positive value of the shock variable corresponds to an unexpectedly tighter monetary policy stance.

The interaction term is negative and statistically significant in both columns, indicating that the automation discount on lending costs widens during episodes of unexpected monetary tightening. This finding can be partly explained by the bank lending channel of monetary policy. When credit conditions tighten unexpectedly, lenders face higher funding costs and reduced risk appetite, leading to sharper discrimination across borrowers on the basis of credit quality. In this environment, the risk signal embedded in automation investment becomes more valuable. Therefore, automated firms seem to benefit disproportionately as lenders reallocate credit among borrowers. In the Turkish context, where the policy rate rose sharply to 50% over the 2023–2024 tightening cycle, the amplification of the automation discount during unexpected tightening episodes carries particular economic relevance.

4.6 Robustness Checks

To assess the sensitivity of the baseline findings, we subject the main specification to a battery of robustness checks, reported in Table 7. Each row corresponds to a separate regression in which we modify one dimension of the baseline model while holding all others constant.

The first set of checks concerns the definition of the treatment variable. Rows (1) and (2) replace the import-based automation indicator with alternative proxies. Row (1) defines automation through the labor force: a firm is classified as automated if it employs

any workers in occupation codes related to automation activities, as recorded in the Social Security Institution employee registry in a given month. The automation-related ISCO-4 codes are 2511, 2523, 3511 and 3512, which correspond to system analysts, computer network professionals, ICT operations technicians and ICT user support technicians, respectively. Row (2) uses a collapsed indicator that equals one if either the import-based or the employment-based measure is positive. The negative and significant coefficient across both specifications confirms that the baseline result does not depend on how automation is operationalized and is detectable through entirely different data sources.

Rows (3) through (7) test alternative constructions of the import-based automation variable. Row (3) replaces the 24-month trailing sum with a permanent treatment indicator that switches to one at the firm's first automation-related import and remains at one thereafter, testing whether the effect reflects a permanent shift in firm status rather than a rolling window. Rows (4) and (5) replace the 24-month window with 12-month and 36-month alternatives, examining sensitivity to the horizon over which automation activity is measured. Rows (6) and (7) replace the binary indicator with continuous measures — the log-transformed value of automation imports and the IHS-transformed share of automation-related imports in total imports, respectively — allowing for intensity of adoption rather than a simple presence indicator. The estimates remain negative and statistically significant throughout, with magnitudes broadly consistent with the baseline.

Rows (8) and (9) address inference concerns by re-estimating the baseline model under alternative standard error clustering strategies. The baseline clusters at the firm level; rows (8) and (9) cluster at the industry-by-time and firm-by-bank levels, respectively. Both alternatives yield tighter standard errors than the baseline, confirming that firm-level clustering is conservative. Rows (10) and (11) assess whether the result is driven by very large firms that both automate systematically and enjoy favourable credit pricing due to size rather than automation per se. We exclude the top 1% of the asset distribution, defined within each year in row (10) and within each year-by-sector cell in row (11). The automation coefficient remains negative and significant in both cases, ruling out this concern.

The final set of checks examines specific sample composition issues. Row (12) aug-

ments the baseline with log-transformed value of firm age and TL-denominated credit share at firm-bank-month level as additional controls, addressing the possibility that younger and more dynamic firms are disproportionately represented among automators and benefit from age-related lending advantages independently of automation. Row (13) excludes firms that utilise rediscount credits, a subsidised lending facility that could introduce non-market pricing into the outcome variable. Row (14) excludes sectors receiving the largest share of investment credits, where directed lending programmes may similarly distort interest rate observations. The automation discount survives all three sample restrictions.

Across all fourteen specifications, the coefficient on *Automation* retains its significance, providing strong evidence that the baseline finding is not an artefact of a particular measurement choice, estimation assumption, or sample composition.

4.7 Other Lending Outcomes

Table 8 extends the baseline analysis beyond the cost of lending to examine whether automation also shapes non-price credit outcomes. Columns (1) and (2) report results for the non-performing loan indicator, while columns (3) and (4) examine loan maturity.

The coefficient on *Automation* is negative and statistically significant in columns (1) and (2), indicating that automated firms are less likely to have a non-performing loan in a given year. This finding complements the interest rate results in two ways. First, it provides direct evidence that the lower lending costs extended to automated firms reflect a genuine reduction in credit risk rather than a mispricing by lenders. Second, it suggests that the operational improvements associated with automation — more stable cash flows, reduced cost structure volatility, and stronger overall financial performance — translate into materially lower default incidence over time, validating the risk-reduction channel discussed earlier.

Turning to loan maturity in columns (3) and (4), the coefficient on *Automation* is positive but only marginally significant, and the result is not robust to the inclusion of firm-level controls. The evidence on maturity is therefore inconclusive, and we refrain from drawing strong inferences from this outcome.

Overall, the results stemming from this analysis suggest that the financial benefits of automation extend beyond interest rate reductions, with automated firms displaying lower credit risk as measured by non-performing loan incidence, while the evidence on maturity remains limited.

4.8 AI Intensity and Complementarity with Automation

In the following stage, we explore whether firm-level adoption of AI-related processes creates any complementary impacts on the potential reduction in cost of credits. In particular, Table 9 examines the role of AI adoption in shaping the cost of lending and its interaction with firm automation. Following the same approach used to construct the baseline automation measure, we define *AI* as a binary indicator taking the value of one if the 24-month trailing sum of a firm's imports of AI-related capital goods — identified through a curated set of 6-digit HS codes covering AI-specific hardware is positive, and zero otherwise. These HS codes cover 847141–847150, 847321, 847330, 847010–847029, which specifically include automatic data processing and electronic calculating machines such as graphics processing units, AI servers, and specialized processors. This approach captures firms that have made tangible, hardware-level investments in AI infrastructure, as distinct from software-based or organizational AI adoption that is not observable in customs records.

Columns (1) and (2) estimate the standalone effect of AI adoption on lending costs. The coefficient on *AI* is negative and marginally significant, suggesting that AI-adopting firms access bank credit at somewhat lower interest rates. Columns (3) and (4) introduce both *Automation* and *AI* alongside their interaction. The coefficient on *Automation* remains negative and significant, consistent with the baseline results. The coefficient on *AI* alone becomes positive and statistically insignificant, while the interaction term $Automation \times AI$ is negative and highly significant. This pattern indicates that the combined effect of automation and AI adoption on lending costs exceeds the sum of their individual contributions — a finding consistent with genuine complementarity between the two technologies in the eyes of lenders.

The complementarity result is economically intuitive. Automated production processes generate structured operational data at high frequency; AI systems are particularly well-suited to process this data in real time, optimizing production scheduling, predicting equipment failures, and reducing waste. A firm deploying both technologies credibly signals a higher level of organizational capability than one deploying either in isolation, and the combined risk reduction exceeds what either technology alone can achieve. This is consistent with the broader evidence that AI investments are associated with improvements in firm-level operational performance and product innovation, and that their productivity effects are amplified when complementary technologies and organizational capabilities are already in place (Babina et al., 2024; Brynjolfsson et al., 2025). From a lender's perspective, a firm combining automation with AI investment represents a more transparent and predictable borrower, whose risk profile is credibly lower than what conventional financial metrics alone would suggest.

4.9 Real Effects of Firm Automation

The preceding analysis establishes that firm automation reduces the cost of bank financing and improves non-price lending outcomes. A natural complementary question is whether automation also shapes the real-side performance of firms — their sales, employment, export activity, productivity, and investment behavior. While the credit market effects documented above speak to how lenders perceive and price automated firms, the real-side analysis provides a broader picture of how automation reshapes firm-level economic outcomes beyond the financing dimension. It also serves as an implicit validation of the credit market findings. In this context, if automated firms genuinely exhibit superior operational performance, the lower borrowing costs they receive are consistent with rational risk pricing rather than a systematic mis-evaluation by lenders.

To examine these real effects, we estimate the following specification at the firm-year level:

$$y_{f,t} = \beta\, Automation_{f,t} + \mathbf{X}'_{f,t}\boldsymbol{\gamma} + \alpha_f + \mu_{s,t} + \varepsilon_{f,t} \tag{3}$$

where $y_{f,t}$ denotes the outcome variable of interest for firm f in year t, and $Automation_{f,t}$ is the binary automation indicator defined as in the baseline credit market specification. $\mathbf{X}_{f,t}$ is the vector of time-varying firm-level controls used throughout the analysis. α_f denotes firm fixed effects and $\mu_{s,t}$ denotes industry-by-year fixed effects, which absorb sector-level demand and productivity cycles that could confound the automation coefficient. Standard errors are clustered at the firm level. The shift from the bank-firm-month panel to a firm-year panel reflects the annual frequency of the financial statement and employment data used to construct the real-side outcome variables.

Table 10 presents the results. The coefficient on *Automation* is positive and statistically significant across all five dependent variables. Automated firms record higher net sales and employment relative to comparable non-automated firms operating in the same industry in the same year, consistent with automation expanding firm productive capacity along both output and labor dimensions. The positive and highly significant coefficient in the export column indicates that automated firms are more likely to participate in international markets and to export larger volumes, suggesting that automation confers competitive advantages that extend beyond the domestic market. The productivity result — positive and marginally significant — points to higher output per worker among automated firms, though the limited precision of this estimate warrants some caution in interpretation. Finally, the investment coefficient is positive and significant, indicating that automated firms allocate a larger share of their asset base to capital expenditures, consistent with automation being part of a broader and sustained commitment to technology-intensive investment strategies rather than a one-off capital goods purchase.

In sum, the real-side evidence shows that automated firms perform better across several operational dimensions. These results are consistent with the view that the lending discount reflects differences in firms' underlying performance and risk profiles.

5 Conclusion

This paper studies whether firm automation is reflected in bank lending terms. Using transaction-level customs records on automation-related capital goods imports linked to

administrative credit registry data for Türkiye, we provide micro-level evidence that automation is systematically reflected in banks' lending decisions. The results show that automated firms obtain bank credit at significantly lower interest rates and also secure larger loan volumes. These findings suggest that automation is not only a production-side investment that enhances operational capacity, but also a financially relevant signal that affects how banks assess and price borrower risk.

The evidence is robust across alternative empirical designs and specifications. The favorable effects of automation remain robust after matching on pre-sample firm characteristics, an event-study design around the introduction of ChatGPT-4, alternative definitions of automation, different clustering choices, and several sample restrictions. Additional results further show that automated firms' loans are less likely to be classified as non-performing, indicating that their lower borrowing costs are consistent with realized improvements in credit risk rather than preferential pricing.

The heterogeneity analyses shed light on the mechanisms underlying this relationship. The discount is not evident for state-owned banks but stronger at banks with greater ICT-related human capital. This result suggests that technologically sophisticated lenders are more responsive to the information contained in automation investment. On the firm side, the effect is stronger among firms with higher sales volatility, deeper banking relationships, and greater agency conflicts. These patterns are consistent with automation reducing uncertainty, improving the borrower-lender information environment, and serving as a credible signal of productive investment.

The findings also point to broader macro-financial relevance. The automation discount widens during episodes of unexpected monetary tightening, when banks screen borrowers more intensively and borrower-quality signals become more valuable. Moreover, physical automation and AI adoption appear to complement each other in reducing lending costs, suggesting that the financial benefits of technological upgrading are strongest when firms combine different layers of digital and production capabilities. Beyond credit markets, automated firms also display stronger real-side performance in terms of sales, employment, exports, productivity, and investment.

These findings carry important implications for policymakers and regulators. Policies that encourage technology adoption may generate benefits that extend beyond productivity gains by improving firms' access to external finance, particularly in bank-dependent emerging markets. At the same time, the results highlight the role of banks' own technological capabilities in translating firm-level innovation into efficient credit allocation. As automation and AI become increasingly embedded in production processes, lenders may increasingly incorporate technology adoption into credit risk assessment frameworks.

Taken together, our results suggest that automation can affect firm outcomes through two reinforcing channels: directly, by improving operational performance, and indirectly, by lowering the cost of external finance. By influencing both productivity and access to credit, automation can play an important role in shaping firm growth and competitiveness. More broadly, the results suggest that technological upgrading may affect not only firm productivity, but also the allocation and pricing of bank credit in emerging markets.

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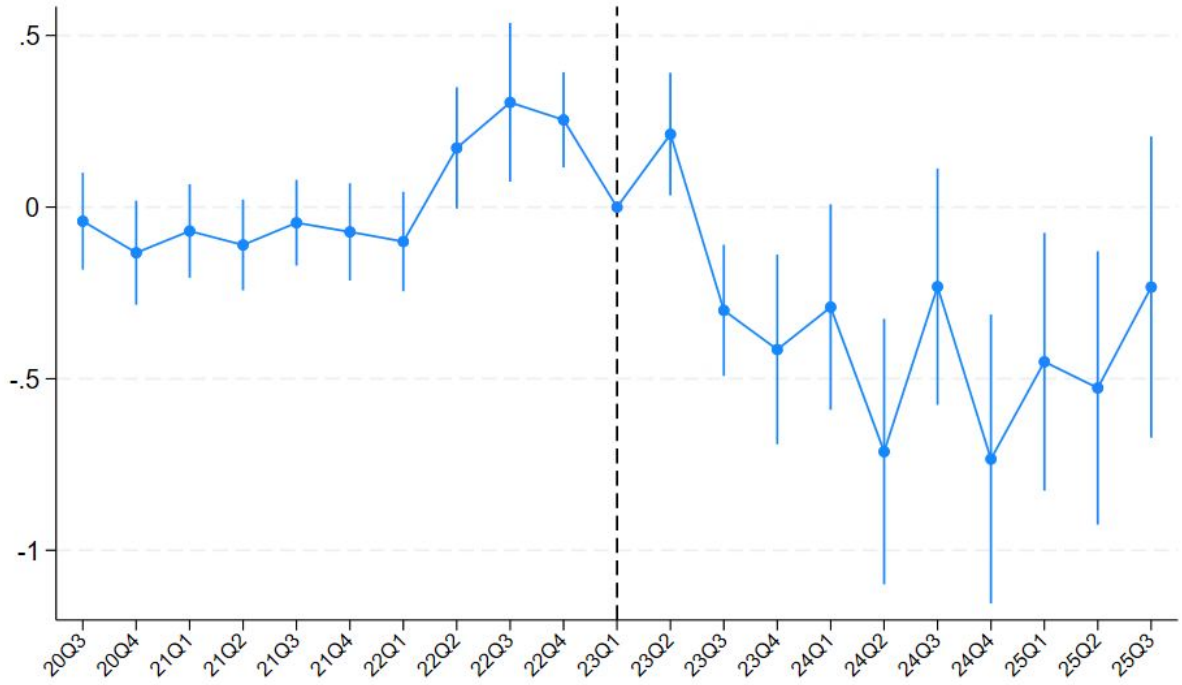
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Figure 1: Event Study: Introduction of ChatGPT-4



Notes: This figure presents the results of an event study examining the differential effect of the introduction of ChatGPT-4 in March 2023 on the cost of lending for automated versus non-automated firms. The sample is the matched sample described in Section 4.2. The model is estimated at monthly frequency over the full sample period, with the *Automation indicator* interacted with a set of relative-time dummies defined at quarterly frequency over a [-10, +10] quarter window around the event date. The dependent variable is *Int Rate*, the weighted average interest rate on TL-denominated loans. All specifications include firm, bank-by-time, and industry-by-time fixed effects and time varying controls. The figure plots the point estimates on the interaction coefficients along with their 95% confidence intervals. Standard errors are clustered at the firm level. The omitted category is the quarter immediately preceding the event date.

Table 1: Summary Statistics

	(1) Obs.	(2) Mean	(3) Std. Dev.	(4) p5	(5) p95
Firm-Bank Level					
Int Rate	2,355,687	28.406	16.284	10.000	59.748
Loans	2,355,687	7.308	1.800	4.331	10.198
Maturity	2,327,198	2.228	0.830	0.642	3.624
Loan Density	2,355,687	38.119	32.525	2.542	100
NPL	790,608	0.001	0.030	0	0
Firm Level					
Automation	1,437,727	0.065	0.246	0	1
AI	1,437,727	0.013	0.114	0	0
Firm Size	236,755	16.114	1.693	13.556	19.143
Profitability	236,755	3.950	7.232	-2.572	18.930
Leverage	236,755	68.090	21.663	25.367	95.379
Liquidity	236,755	190.802	276.999	67.828	465.903
Tangibility	236,755	26.672	19.808	2.277	65.872
Sales Volatility	1,422,761	0.978	6.062	0.005	3.952
Agency Conflicts	297,977	0.139	0.342	0	1
Sales	236,755	16.501	1.852	13.612	19.631
Employment	236,755	3.033	1.215	1.386	5.278
Exports	236,755	6.919	7.574	0	17.977
Productivity	236,755	13.565	1.278	11.611	15.680
Investment	236,755	36.180	85.963	0	136.422
Bank Level					
State Bank	2,870	0.232	0.422	0	1
Bank ICT Emp	2,726	1.042	1.951	0	4.021

Notes: This table presents the summary statistics of the variables used in the analyses. Column (1) gives the number of observations. Columns (2), (3), (4) and (5) provide the mean, standard deviation, 5th percentile and 95th percentile values, respectively.

Table 2: Baseline Results

	(1) Int Rate	(2) Int Rate	(3) Loans	(4) Loans
Automation	-0.084***	-0.090***	0.068***	0.033***
	(0.025)	(0.024)	(0.008)	(0.007)
Obs.	2,432,783	2,355,687	2,432,783	2,355,687
Firm FE	✓	✓	✓	✓
Bank x Time FE	✓	✓	✓	✓
Industry x Time FE	✓	✓	✓	✓
Firm Controls		✓		✓
Adj. R^2	0.951	0.952	0.712	0.699

Notes: This table reports the baseline estimates of the effect of firm automation on the cost of lending and loan volume. The dependent variable in columns (1) and (2) is *Int Rate*, the weighted average interest rate on Turkish lira-denominated loans. The dependent variable in columns (3) and (4) is *Loans*, the log-transformed TL-denominated credit balance. The key independent variable, *Automation*, is a binary indicator taking the value of one if the 24-month trailing sum of a firm's automation-related capital goods imports is positive, and zero otherwise. All specifications include firm, bank-by-time, and industry-by-time fixed effects. Columns (2) and (4) additionally include time-varying firm-level controls comprising firm size, profitability, leverage, liquidity, and tangibility. Standard errors are clustered at the firm level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 3: Alternative Empirical Design

	(1) Int Rate	(2) Int Rate	(3) Int Rate	(4) Int Rate
Automation	-0.071*	-0.092**	0.031	0.004
	(0.042)	(0.043)	(0.044)	(0.044)
Automation x Post ChatGPT-4			-0.308***	-0.292***
			(0.084)	(0.086)
Obs.	290,786	271,282	290,786	271,282
Sample	Matched	Matched	Matched	Matched
Firm FE	✓	✓	✓	✓
Bank x Time FE	✓	✓	✓	✓
Industry x Time FE	✓	✓	✓	✓
Firm Controls		✓		✓
Adj. R^2	0.944	0.945	0.944	0.945

Notes: This table reports estimates of the effect of firm automation on the cost of lending under alternative identification strategies. All columns use a matched sample constructed via 1-to-1 multivariate distance matching without replacement, based on firm-level controls with exact matching on 2-digit NACE sector classifications and regulatory firm size categories, yielding 3,726 treated and 3,726 control firms. Columns (1) and (2) repeat the baseline specification on the matched sample. Columns (3) and (4) additionally introduce *Automation* × *Post ChatGPT-4*, exploiting the exogenous introduction of ChatGPT-4 in March 2023 as a shock to automation-complementary technologies. The dependent variable in all columns is *Int Rate*. All specifications include firm, bank-by-time, and industry-by-time fixed effects. Columns (2) and (4) additionally include firm-level controls. Standard errors are clustered at the firm level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 4: Bank Heterogeneity

	(1) Int Rate	(2) Int Rate
Automation	-0.313***	-0.042
	(0.028)	(0.027)
Automation x State Bank	0.529***	
	(0.033)	
Automation x Bank ICT Emp		-0.009***
		(0.002)
Obs.	2,355,687	2,134,371
Firm FE	✓	✓
Bank x Time FE	✓	✓
Industry x Time FE	✓	✓
Firm Controls	✓	✓
Adj. R^2	0.952	0.951

Notes: This table reports estimates of the effect of firm automation on the cost of lending across different bank types. The dependent variable in all columns is *Int Rate*. Column (1) introduces *Automation* × *State Bank*, an interaction between the *Automation* indicator and a binary variable taking the value of one for state-owned banks. Column (2) introduces *Automation* × *Bank ICT Emp*, an interaction between the *Automation* indicator and the share of ICT-related employees in the bank's total workforce. All specifications include firm, bank-by-time, and industry-by-time fixed effects and time-varying firm-level controls. Standard errors are clustered at the firm level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 5: Firm Heterogeneity

	(1) Int Rate	(2) Int Rate	(3) Int Rate
Automation	-0.062**	-0.048	-0.067***
	(0.024)	(0.029)	(0.024)
Automation x Sales Volatility	-0.010***		
	(0.003)		
Automation x Loan Density		-0.002**	
		(0.001)	
Automation x Agency Conflicts			-0.162***
			(0.047)
Sales Volatility	-0.003**		
	(0.002)		
Loan Density		-0.010***	
		(0.000)	
Agency Conflicts			-0.016
			(0.013)
Obs.	2,344,860	2,355,687	2,355,687
Firm FE	✓	✓	✓
Bank x Time FE	✓	✓	✓
Industry x Time FE	✓	✓	✓
Firm Controls	✓	✓	✓
Adj. R^2	0.952	0.952	0.952

Notes: This table reports estimates of the effect of firm automation on the cost of lending across different firm characteristics. The dependent variable in all columns is *Int Rate*. Column (1) introduces *Automation* × *Sales Volatility*, an interaction between the *Automation* indicator and the 12-month standard deviation of gross sales. Column (2) introduces *Automation* × *Loan Density*, an interaction between the *Automation* indicator and the share of a given bank in the firm's total TL-denominated credit borrowing. Column (3) introduces *Automation* × *Agency Conflicts*, an interaction between the *Automation* indicator and a binary variable taking the value of one for firms with an above-median free cash flow to assets ratio and a below-median asset growth rate. All specifications include firm, bank-by-time, and industry-by-time fixed effects and time-varying firm-level controls. Standard errors are clustered at the firm level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 6: Monetary Policy Shock

	(1) Int Rate	(2) Int Rate
Automation	-0.099***	-0.101***
	(0.025)	(0.024)
Automation x MP Shock	-0.092***	-0.071***
	(0.006)	(0.006)
Obs.	2,432,783	2,355,687
Firm FE	✓	✓
Bank x Time FE	✓	✓
Industry x Time FE	✓	✓
Firm Controls		✓
Adj. R^2	0.951	0.952

Notes: This table reports estimates of the effect of firm automation on the cost of lending across different monetary policy conditions. The dependent variable in all columns is *Int Rate*. *Automation* × *MP Shock* is an interaction between the *Automation* indicator and a monetary policy shock variable constructed as the residual from a regression of the difference between the actual policy rate and market expectations — sourced from the CBRT Survey of Market Participants —on a set of macroeconomic predictors, including 12-month forward inflation expectation, annual change in industrial production index, annual credit growth and annual change in Istanbul Stock Exchange (BIST100) index. A positive value of the shock variable corresponds to an unexpectedly tighter monetary policy stance. All specifications include firm, bank-by-time, and industry-by-time fixed effects. Column (2) additionally includes time-varying firm-level controls. Standard errors are clustered at the firm level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 7: Robustness Checks

	(1) Coefficient (β)	(2) SE
(1) Defining Automation with employment	-0.099***	(0.046)
(2) Defining Automation with imports and employment	-0.068***	(0.023)
(3) Permanent treatment Automation indicator	-0.085***	(0.024)
(4) 12-month window Automation	-0.089***	(0.024)
(5) 36-month window Automation	-0.073***	(0.024)
(6) Automation imports amount as treatment	-0.009***	(0.002)
(7) Automation imports share as treatment	-0.018**	(0.008)
(8) SEs clustered at sector-time level	-0.090***	(0.017)
(9) SEs clustered at bank-firm level	-0.090***	(0.019)
(10) Excluding largest firms (year level)	-0.085***	(0.024)
(11) Excluding largest firms (year-sector level)	-0.099***	(0.024)
(12) Controlling for firm age and TL credit share	-0.076***	(0.025)
(13) Excluding firms using rediscount credits	-0.073***	(0.021)
(14) Excluding sectors receiving largest investment credits	-0.129***	(0.029)

Notes: This table reports robustness checks for the baseline estimate of the effect of firm automation on the cost of lending. Each row corresponds to a separate regression where the dependent variable is *Int Rate* and all specifications include firm, bank-by-time, and industry-by-time fixed effects and time-varying firm-level controls, unless otherwise stated. Column (1) reports the coefficient on the *Automation* indicator and column (2) reports the corresponding standard error. Rows (1) and (2) test alternative automation definitions: row (1) defines automation as a binary indicator taking the value of one if the firm employs any workers in occupation codes related to automation activities, and zero otherwise, while row (2) uses a collapsed indicator taking the value of one if either the import-based or the employment-based measure is positive. Row (3) uses a permanent treatment indicator that switches to one upon the firm's first automation-related import and remains one thereafter. Rows (4) and (5) replace the baseline 24-month trailing sum window with 12-month and 36-month windows, respectively. Rows (6) and (7) replace the binary automation indicator with the log-transformed value and the IHS-transformed share of automation-related imports in total imports, respectively. Rows (8) and (9) re-estimate the baseline model clustering standard errors at the industry-by-time and bank-by-firm levels, respectively, instead of the baseline firm level. Rows (10) and (11) exclude the largest firms based on the top 1% of the asset distribution within each year and within each year-by-sector cell, respectively. Row (12) augments the baseline specification with log-transformed value of firm age and TL-denominated credit share in total credits at firm-bank-month level as additional control variables. Row (13) excludes firms that utilize rediscount credits. Row (14) excludes sectors receiving the largest share of investment credits. Standard errors are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 8: Other Lending Outcomes

	(1) NPL	(2) NPL	(3) Maturity	(4) Maturity
Automation	-0.0009**	-0.0010***	0.006	0.007*
	(0.0004)	(0.0004)	(0.004)	(0.004)
Obs.	817,902	790,608	2,403,673	2,327,198
Time Frequency	Annual	Annual	Monthly	Monthly
Firm FE	✓	✓	✓	✓
Bank x Time FE	✓	✓	✓	✓
Industry x Time FE	✓	✓	✓	✓
Firm Controls		✓		✓
Adj. R^2	0.423	0.436	0.443	0.444

Notes: This table reports estimates of the effect of firm automation on non-price lending outcomes. The dependent variable in columns (1) and (2) is *NPL*, a binary indicator taking the value of one if the firm has any non-performing loan in a given year, and zero otherwise. The dependent variable in columns (3) and (4) is *Maturity*, the log-transformed number of months to the maturity of credits. The observations of maturity with zero-month are excluded. The key independent variable, *Automation*, is a binary indicator taking the value of one if the 24-month trailing sum of a firm's automation-related capital goods imports is positive, and zero otherwise. All specifications include firm, bank-by-time, and industry-by-time fixed effects. Columns (2) and (4) additionally include time-varying firm-level controls comprising firm size, profitability, leverage, liquidity, and tangibility. Standard errors are clustered at the firm level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 9: Results for AI Adoption

	(1) Int Rate	(2) Int Rate	(3) Int Rate	(4) Int Rate
AI	-0.101*	-0.102*	0.076	0.059
	(0.054)	(0.056)	(0.074)	(0.076)
Automation x AI			-0.248***	-0.243***
			(0.089)	(0.090)
Automation			-0.066***	-0.073***
			(0.025)	(0.024)
Obs.	2,432,783	2,355,687	2,432,783	2,355,687
Firm FE	✓	✓	✓	✓
Bank x Time FE	✓	✓	✓	✓
Industry x Time FE	✓	✓	✓	✓
Firm Controls		✓		✓
Adj. R^2	0.951	0.952	0.951	0.952

Notes: This table reports estimates of the effect of AI adoption on the cost of lending and its complementarity with firm automation. The dependent variable in all columns is *Int Rate*. *AI* is a binary indicator taking the value of one if the 24-month trailing sum of a firm's AI-related capital goods imports is positive, and zero otherwise. Columns (1) and (2) estimate the standalone effect of AI adoption on lending costs. Columns (3) and (4) additionally introduce *Automation* and its interaction with *AI* to examine whether physical automation and AI adoption exhibit complementarities in shaping the cost of lending. All specifications include firm, bank-by-time, and industry-by-time fixed effects. Columns (2) and (4) additionally include time-varying firm-level controls comprising firm size, profitability, leverage, liquidity, and tangibility. Standard errors are clustered at the firm level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 10: Real Effects of Firm Automation

	(1) Sales	(2) Employment	(3) Exports	(4) Productivity	(5) Investment
Automation	0.087***	0.064***	0.311***	0.021**	5.463***
	(0.006)	(0.004)	(0.050)	(0.005)	(0.775)
Obs.	230,591	230,591	230,591	230,591	230,295
Firm FE	✓	✓	✓	✓	✓
Industry x Year FE	✓	✓	✓	✓	✓
Firm Controls	✓	✓	✓	✓	✓
Adj. R^2	0.931	0.944	0.790	0.873	0.337

Notes: This table reports estimates of the effect of firm automation on real-side firm outcomes. Unlike the baseline specifications, the regressions in this table are estimated at the firm-year level and include firm and industry-by-year fixed effects. We use 12-month trailing sum to specify automated firms in order to be compatible with firm-year level data. The dependent variable in column (1) is *Sales*, the log-transformed annual net sales. The dependent variable in column (2) is *Employment*, the log-transformed annual average number of employees. The dependent variable in column (3) is *Exports*, the log-transformed annual total firm exports. The dependent variable in column (4) is *Productivity*, the log-transformed annual net sales per employee. The dependent variable in column (5) is *Investment*, the ratio of current year capital expenditures to previous year total assets. The key independent variable, *Automation*, is a binary indicator taking the value of one if the 24-month trailing sum of a firm's automation-related capital goods imports is positive, and zero otherwise. All specifications include firm and industry-by-year fixed effects and time-varying firm-level controls comprising firm size, profitability, leverage, liquidity, and tangibility. Standard errors are clustered at the firm level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Appendix

Table A1: Variable Definitions

	(1) Definition	(2) Source
Int Rate	The weighted average interest rate applicable to credits	Credit Registry
Loans	The log-transformed TL-denominated credit balance	Credit Registry
Maturity	The log-transformed number of months to the maturity of credits	Credit Registry
NPL	A binary indicator taking the value of one if a firm has a non-performing loan in a year, otherwise zero	Credit Registry
Automation	A binary indicator taking the value of one if 24 months trailing sum of automation-related capital goods imports is positive, otherwise zero	Ministry of Trade, Acemoglu et al. (2020)
AI	A binary indicator taking the value of one if 24 months trailing sum of AI-related capital goods imports is positive, otherwise zero	Ministry of Trade, Acemoglu et al. (2020)
Firm Size	The log-transformed total assets	Revenue Administration
Profitability	The ratio of net income to total assets	Revenue Administration
Leverage	The ratio of total liabilities to total assets	Revenue Administration
Liquidity	The ratio of current assets to current liabilities	Revenue Administration
Tangibility	The ratio of tangible assets to total assets	Revenue Administration

Notes: This table presents the definitions and data sources of the variables used in the analyses. Column (1) gives the detailed explanations regarding the definitions of the variables, whereas column (2) provides the information about data sources.

Table A2: Variable Definitions (contd)

	(1) Definition	(2) Source
Sales Volatility	12 months standard deviation of gross sales	Revenue Administration
Loan Density	The share of a given bank in the firm's total TL-denominated credit borrowing	Credit Registry, Revenue Administration
Agency Conflicts	A binary indicator taking the value of one if FCF to assets ratio is higher than median value and asset growth rate is lower than median value, otherwise zero	Revenue Administration
Sales	The log-transformed annual net sales	Revenue Administration
Employment	The log-transformed annual average number of employees	Social Security Institution
Exports	The log-transformed annual total firm exports	Ministry of Trade
Productivity	The log-transformed annual net sales per employee	Revenue Administration, Social Security Institution
Investment	The ratio of current year capital expenditures to previous year assets	Revenue Administration
State Bank	A binary indicator taking the value of one for state-owned banks, otherwise zero	BRSA
Bank ICT Emp	The ratio of ICT-related employees to total employees of banks	BRSA, Social Security Institution

Notes: This table presents the definitions and data sources of the variables used in the analyses. Column (1) gives the detailed explanations regarding the definitions of the variables, whereas column (2) provides the information about data sources.